Rice Leaf Disease Detection in the Philippines Using EfficientNet-Based Convolutional Neural Networks with Multi-Metric Performance Evaluation

Francine Angela G. Alejandro
School of Computing,
Holy Angel University
Angeles City, Pampanga, Philippines
+639287207896
cinella.alejandro07@gmail.com

Yohanna A. Decilio
School of Computing,
Holy Angel University
Angeles City, Pampanga, Philippines
+639629393821
yohannadecilio@gmail.com

Shane S. Mendoza
School of Computing
Holy Angel University
Angeles City, Pampanga, Philippines
+639310482585
mendozashane0820@gmail.com

Maria Niña Grace L. Pascua
School of Computing
Holy Angel University
Angeles City, Pampanga, Philippines
+639150547753
ninapascua88@gmail.com

ABSTRACT

Rice is a staple crop that plays a crucial role in national food security and agricultural livelihoods, particularly in the Philippines. However, rice production is often affected by leaf diseases such as Leaf Blast, Brown Spot, and Rice Hispa, which can significantly reduce crop yields by 20%–70% during severe outbreaks. Identifying these diseases typically relies on the expertise and observation of farmers and agricultural specialists. While this expertise is valuable, timely and large-scale screening can still be difficult, especially when symptoms appear similar across diseases, early signs are subtle, or expert support is not immediately available.

This project proposes a rice leaf disease detection system using an EfficientNet-based Convolutional Neural Network (CNN) for image classification. The system allows users to upload rice leaf images and obtain the predicted disease, its confidence score, and a short explanation of the result. Model performance was evaluated using accuracy and macro-F1 score, while the per-class confusion matrix was used for class-level error analysis, and Grad-CAM visualization was used to highlight the image regions influencing model classification and improve interpretability. To improve interpretability, an auxiliary Natural Language Processing (NLP) component retrieves short explanations of the detected disease using TF-IDF similarity matching, and a reinforcement learning (RL) component uses a Q-learning agent to tune the classification confidence threshold based on improvements in validation macro-F1 score. Experimental results indicate that the proposed model achieves high classification performance, with an accuracy of 0.9883 and a macro-F1 score of 0.9827. Ablation experiments further demonstrate that while data augmentation improves generalization, class weighting has a more significant impact on balanced performance, particularly in improving minority class recall.

CCS Concepts

- **Computing Methodologies** → Machine Learning;
- **Applied Computing** → Agriculture;
- **Computing Methodologies** → Neural Networks

Keywords

Rice leaf disease detection; EfficientNetB0; Grad-CAM; NLP advisory system; reinforcement learning; plant disease classification; Pampanga; food security

1. INTRODUCTION

Rice is the primary staple crop of the Philippines and accounts for around 20% of the gross value added of Philippine agriculture [1], while the rice sector supports about 2.5 million households [2]. This importance is also reflected in Pampanga, a major rice-producing province in Central Luzon, where the crop holds deep economic and cultural significance. However, despite its importance, rice leaf diseases remain a persistent threat to yield stability and farmer income.

Experienced Filipino rice farmers possess substantial field-based expertise regarding their land and crops—many can identify disease signs from subtle changes in leaf color, texture, and growth pattern long before visible lesions fully develop. Even so, accurate identification can remain difficult when multiple diseases appear simultaneously or when symptoms present atypically under varying field conditions. Three diseases are particularly prevalent in Philippine paddy fields: Leaf Blast (*Magnaporthe grisea*), Brown Spot (*Bipolaris oryzae*), and Rice Hispa (*Dicladispa armigera*). Collectively, these diseases can substantially reduce rice yield under severe outbreak conditions [3].

The three diseases exhibit strong inter-class visual similarity and wide intra-class variability influenced by cultivar, growth stage, and field environment. Images collected in real paddy fields introduce additional variation from changing light conditions, background clutter, moisture, and co-occurring insect damage [3]. These factors make even well-trained classifiers prone to error in visually ambiguous cases.

This project proposes a rice leaf disease detection system: a fine-tuned EfficientNetB0 convolutional neural network that classifies four classes from field photographs and generates Grad-CAM saliency maps, enabling inspection and verification of each classification. The system also includes an auxiliary Natural

Language Processing (NLP) component that retrieves short explanations describing the predicted rice disease using TF-IDF similarity matching from a small disease knowledge base. In addition, a reinforcement learning component was implemented to tune the classification confidence threshold using a simple Q-learning agent that learns a threshold policy based on improvements in validation macro-F1 score. The rice leaf disease detection system was designed as a decision-support tool that provides additional evidence to farmers, supporting—not replacing—the judgment they have built through years of working the land.

2. PROBLEM STATEMENT AND SCOPE

The study addresses the challenge of accurately and consistently identifying rice leaf diseases under Philippine field conditions, where visually similar symptoms, environmental variation, and limited immediate technical support can make diagnosis difficult. The project aims to develop an EfficientNet-based rice leaf disease detection system that is accurate, interpretable, and practical as a decision support tool for farmers and agricultural specialists.

2.1 Scope

This project focuses on the development of a rice leaf disease detection system that classifies images of rice leaves into four classes: the three common diseases affecting Philippine rice fields (Leaf Blast, Brown Spot, and Rice Hispa) and a Healthy class. The system uses an EfficientNet-based Convolutional Neural Network (CNN) to analyze uploaded leaf images and predict the corresponding disease category with a confidence score.

The study includes dataset preparation, image preprocessing, model training, and evaluation using performance metrics such as accuracy, macro-F1 score, and per-class confusion matrix. Grad-CAM visualization is used to highlight the regions of the leaf image that influenced the model's classification, improving interpretability for users.

In addition to the CNN-based classification component, an auxiliary NLP module retrieves short explanations of the detected disease from a small knowledge base using TF-IDF similarity matching and displays them with the predicted disease and confidence score. For development purposes, model evaluation metrics such as accuracy, macro-F1 score, and the per-class confusion matrix are also used. A reinforcement learning component using a simple Q-learning agent is also implemented to tune the classification confidence threshold based on the validation macro-F1 score.

The system is designed as a decision support tool to assist farmers and agricultural specialists in crop monitoring and disease detection. However, the project is limited to image-based classification and does not include real-time field deployment, treatment recommendation systems, or integration with mobile or agricultural management platforms. Model performance may also be influenced by variations in lighting conditions, leaf appearance, and dataset quality.

2.2 Success Metrics and Constraints

2.2.1 Success Metrics

The performance of the CNN model was evaluated using key classification metrics to assess classification reliability and class-level performance. The primary evaluation metrics include accuracy, macro-F1 score, and per-class confusion matrix. Accuracy measures the overall proportion of correctly classified rice leaf images. The macro-F1 score evaluates model performance across all disease classes by equally considering each class in the dataset, which is important for multiclass disease classification. The per-class confusion matrix provides a detailed breakdown of classification results, allowing analysis of how frequently specific diseases are correctly classified or confused with other classes.

2.2.2 Constraints

Several constraints may influence the implementation of the project. First, the rice leaf dataset used in this study exhibits class imbalance, where some diseases have significantly more samples than others, potentially biasing model learning. Second, the images were collected in controlled environments with clean, uniform backgrounds, meaning the model may struggle when exposed to real-world field images that contain noise such as dirt, clutter, or mixed lighting conditions. Finally, computational limitations and project time constraints may limit the number of model architectures or hyperparameter configurations that can be explored.

3. RELATED WORK

CNNs have established a strong record in plant disease recognition. Mohanty et al. [4] showed that deep networks trained on the PlantVillage dataset could identify 26 diseases across 14 crops with over 99% laboratory accuracy, though performance fell significantly on field images—highlighting the domain gap that motivates the augmentation strategies used in this work. This finding is important because it shows that strong results in controlled datasets do not always guarantee the same level of performance in real agricultural environments. In practical field conditions, leaf images may contain uneven lighting, blurred details, overlapping leaves, and noisy backgrounds, all of which can reduce classification accuracy. As a result, plant disease detection systems must be designed not only for high laboratory performance but also for strong adaptability to real-world conditions. Chin and Suhaili [5] highlight inter-class similarity as a challenge in fine-grained paddy disease classification. This challenge becomes even more significant in rice disease detection because several diseases share similar discoloration patterns, lesion shapes, or early visual symptoms. Such visual overlap makes accurate classification difficult and increases the need for a model that can extract more discriminative and fine-grained image features.

Tan and Le [6] introduced EfficientNet, which applies compound scaling to network depth, width, and input resolution simultaneously. EfficientNetB0 matches the accuracy of much larger models at a fraction of the parameter count, making it well-suited to web or edge deployment scenarios where inference speed and model size matter. This efficiency motivates its selection as the backbone for rice leaf disease detection.

Mumuni and Mumuni [7] surveyed data augmentation strategies, demonstrating that geometric and photometric transforms—rotation, brightness jitter, contrast adjustment, and horizontal flipping—substantially improve generalization when training images are environmentally homogenous. Buda et al. [8] showed that class imbalance suppresses minority-class recall in CNNs and that macro-F1 is a more honest performance indicator for rice leaf disease detection's training and evaluation design.

Sarkar [9] showed that Grad-CAM can improve interpretability in rice disease classification by highlighting lesion regions relevant to the model's prediction. This finding supports Grad-CAM as an interpretability feature in rice leaf disease detection.

The findings of these studies align closely with the characteristics of the dataset used in this project. The RiceLeafs dataset is collected under controlled laboratory-like conditions, featuring clean backgrounds and consistent lighting, similar to the PlantVillage-style images referenced by Mohanty et al. As highlighted in their work, such controlled datasets may lead to models that perform well in training but struggle on real field images—a limitation that this project addresses using augmentation strategies recommended by Mumuni and Mumuni. Additionally, the dataset exhibits class imbalance, directly reflecting the imbalance effects described by Buda et al., reinforcing the decision to evaluate models using macro-F1 and to apply techniques that mitigate skewed class distributions.

4. DATASET PLAN

4.1 Sources and Licensing

The dataset used in this study consists of the Rice Leaf Diseases Detection [10] from Kaggle, which contains images of rice leaf disease classes along with healthy samples, the dataset was filtered to include only the selected four classes. Only this single dataset was used for training and evaluation. All images were filtered, organized, and labeled according to the four target classes defined in the project. The dataset was used in accordance with the license provided on the Kaggle platform. Healthy leaf samples are included for comparison. No identifiable human subjects or private land data are collected or stored.

4.2 Labels and Representativeness

Four (4) classes are defined: Leaf Blast, Brown Spot, Rice Hispa, and Healthy. Although these classes represent common rice leaf conditions found in the literature [3], the dataset used in this study consists of images captured in controlled environments. As a result, factors such as natural lighting changes and field noise are not fully represented in the dataset.

5. METHODOLOGY

5.1 Convolutional Neural Network (CNN) Component

The primary classification component uses EfficientNetB0 [6] with ImageNet pre-trained weights via transfer learning. Data augmentation techniques, including brightness and contrast adjustments, and flipping, are applied [7]. Grad-CAM visual explanations [11] offer interpretability by highlighting the image regions most influential to each prediction. Evaluation metrics include accuracy, macro-F1 score, and per-class confusion matrix.

5.1.1 Training Hyperparameters

The model was trained in two phases using TensorFlow 2.11/Keras on a Windows CPU environment. In the first phase (frozen backbone), the EfficientNetB0 base was kept non-trainable, and the classification head was trained for up to 20 epochs using the Adam Optimizer with a learning rate of $1x10^{-4}$, sparse categorical cross-entropy loss, and a batch size of 32. Early stopping monitored validation accuracy, and ModelCheckpoint saved the best weights. In the second fine-tuning phase, the top 150 layers of the EfficientNetB0 backbone were unfrozen and trained for an additional 10 epochs (epochs 21-30) with a reduced learning rate of $1x10^{-5}$. Class weights were applied throughout both phases. All input images were resized to 224x224 pixels and preprocessed using EfficientNet's built-in preprocess_input normalization. The total model parameter count was 4,059,815, of which 7,684 were trainable during the first phase.

Table 1. Training Configuration Summary

Parameter	Phase 1 (Frozen)	Phase 2 (Fine-Tuning)
Backbone layers	All frozen	Top 150 unfrozen
Trainable params	7,684	~4,059,815
Optimizer	Adam	Adam
Learning rate	$1x10^{-4}$	$1x10^{-5}$
Epochs	Up to 20	21-30
Batch size	32	32
Loss function	Sparse categorical cross entropy	Sparse categorical cross entropy
Class weights	Applied	Applied

5.2 Natural Language Processing Component

An auxiliary Natural Language Processing (NLP) component was implemented to retrieve and generate short explanations describing the detected rice disease. The system uses a small disease knowledge base containing descriptions of symptoms and characteristics for each disease class. After classification, the NLP module retrieves the relevant description using TF-IDF similarity matching and presents it as a concise explanation accompanying the predicted disease and confidence score. For development and evaluation purposes, additional model performance metrics such as accuracy, macro-F1 score, and the per-class confusion matrix were also displayed to provide insight into the system's performance. This allows users to better understand the model's prediction through natural language descriptions. The NLP component was qualitatively evaluated based on the relevance and correctness of the retrieved explanations, ensuring that the generated descriptions align with the predicted disease classes.

Table 2. Sample Output from the System.

Field	Value
True Label	Leaf Blast
Predicted disease	Leaf Blast
Confidence score	0.9911 (99.11%)
TF-IDF retrieval score	0.3476
Retrieved explanation	"*Leaf Blast is the predicted class. Common symptoms include diamond-shaped with gray or whitish centers and brown margins. It is usually related to fungal infection and often spreads under humid*

conditions. A possible next step is to monitor infected plants and consult recommended rice disease management practices."

The TF-IDF retrieval correctly matched the predicted class in all tested cases, confirming that the NLP module consistently retrieves the appropriate disease description. The knowledge base consists of four entries (one per class), each containing symptoms, cause, and suggested action fields.

5.3 Reinforcement Learning Component

A reinforcement learning component was implemented to tune the classification confidence threshold using a simple Q-learning agent [12]. The agent interacts with a validation environment where actions adjust the prediction threshold applied to model outputs. Rewards are computed based on improvements in the macro-F1 score on the validation dataset. Through this interaction, the agent learns a threshold policy that maximizes classification performance.

5.4 Minimum Viable Product (MVP)

The minimum viable product implements the core functionality of the rice leaf disease detection system using the EfficientNetB0-based CNN model for image classification. Users were able to provide rice leaf images through the current notebook-based prototype and receive the predicted disease category along with a confidence score. Grad-CAM visualization highlighted the regions of the leaf image that most influenced the model's prediction, while a short explanation retrieved by the NLP component from a small disease knowledge base described the disease alongside the predicted class and confidence score.

6. DATA PREPARATION & EDA

6.1 Dataset Acquisition Summary

The dataset for this project was obtained from Kaggle, focusing on rice leaf diseases. The dataset contains a total of 5,133 images across four classes, as summarized in Table 1.

Table 3. Dataset Class Distribution

Class	Number of Images
Brown Spot	1480
Leaf Blast	1801
Rice Hispa	1461
Healthy	391

The dataset is imbalanced, with Leaf Blast having the highest number of samples and Healthy having significantly fewer samples compared to other classes. This imbalance may affect model performance and requires handling techniques such as class weighting or data augmentation.

6.2 Train/Validation Split

The dataset was split into training and validation sets. A stratified split was applied to maintain class distribution across both sets. Care was taken to ensure that there was no data leakage between the training and validation sets, maintaining the integrity of model evaluation.

Table 4. Train/Validation Split

Class	Train	Validation
Brown Spot	1184	296
Leaf Blast	1440	361
Rice Hispa	1169	292
Healthy	313	78
Total	4106	1027

Table 4 presents the number of images allocated to the training and validation sets for each class. A stratified split was applied to ensure that the class proportions were preserved across both subsets. This approach supports reliable model evaluation while reducing the risk of sampling bias.

6.3 Exploratory Data Analysis (EDA)

6.3.1 Sample Images per Class

Representative images from each class were examined to understand disease characteristics and visual patterns. Figure 1 shows sample images for all four classes: Brown Spot, Leaf Blast, Rice Hispa, and Healthy.

Brown Spot Leaf Blast Rice Hispa Healthy

Figure 1. Sample images of rice leaves for each class

Each class exhibits distinct visual features:

- **Brown Spot:** Circular brown lesions scattered across the leaf surface.
- **Leaf Blast**: Elongated lesions with grayish centers and dark borders
- **Rice Hispa:** Linear scraping damage and streak-like patterns on leaves
- **Healthy:** Uniform green coloration with no visible disease symptoms

These differences suggest that the classes are visually separable, making them suitable for image-based classification using deep learning models.

6.3.2 Image Resolution Distribution

Image dimensions were analyzed to understand variability in the dataset. Images were resized to 224x224 pixels before model training to ensure consistency and compatibility with the EfficientNet-B0 architecture.

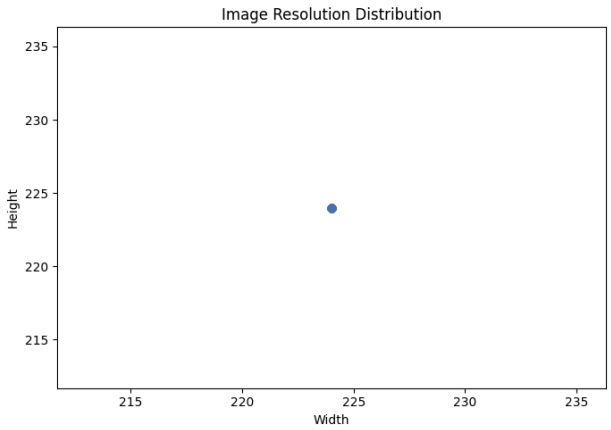

Figure 2. Image Resolution Distribution Graph

Figure 2 illustrates that all images were standardized to an input size of 224 x 224 pixels after preprocessing. This standardization ensures compatibility with the EfficientNetB0 architecture, improves training stability, reduces computational complexity, and supports consistent feature extraction

6.3.3 Color and Brightness Patterns

Disease classes exhibit distinct color and texture characteristics:

- **Brown Spot:** Brown lesions are often surrounded by yellow halos.
- **Leaf Blast:** Grayish or light brown lesions with darker edges.
- **Rice Hispa:** Pale streaks and scraping marks caused by insect feeding.

Healthy leaves are mostly green with uniform brightness. Although slight variations in lighting and orientation exist, most images were captured under controlled conditions, resulting in relatively uniform brightness and background.

6.3.4 Class Imbalance

The dataset shows an uneven distribution across classes, with Leaf Blast having the highest number of samples (1801) and Healthy having the lowest (391). This class imbalance may lead to model bias toward majority classes, potentially reducing performance on underrepresented classes. To address this issue, class weights were applied during training, assigning higher importance to minority classes within the loss function to ensure more balanced learning.

Table 5. Class Weights Used for Training the Model

Class	Label	Weight
Brown Spot	0	0.8670
Healthy	1	3.2796
Leaf Blast	2	0.7128
Hispa	3	0.8781

The healthy class has the highest weight, reflecting its smaller sample size and ensuring the model does not neglect it during training.

6.3.5 Dataset Limitations

- Images were captured under mixed conditions, where some have controlled (white/plain) backgrounds, while others contain natural and varied backgrounds.
- Leaves are clean and well-isolated, which may not reflect real-world field conditions.
- Variations in lighting, background, and image quality exist but are not highly diverse.
- Model performance may vary when applied to images with complex backgrounds, occlusions, or environmental noise.

The EDA confirms that while disease patterns are visually distinct and suitable for classification, dataset imbalance and controlled capture conditions may influence model performance. Addressing these factors through class weighting and data augmentation is essential to improve model robustness and real-world applicability.

7. BASELINE MODELS & INITIAL RESULTS

7.1 Baseline Machine Learning Models

To provide reference performance for the rice leaf disease dataset, several simple machine learning models were implemented. The models include:

- **Logistic Regression** – trained using handcrafted features extracted from the images (e.g., color histograms, texture features).
- **Simple CNN from scratch** – a small convolutional neural network trained directly on the dataset without pretrained weights. The architecture consists of a few convolutional and pooling layers followed by fully connected layers with a softmax output.

These baseline models serve as reference points to evaluate the dataset's complexity and identify classes that are challenging to classify. Figure 3 shows the training and validation accuracy curves of the Simple CNN baseline model during its initial training stage. The overall and class-level results of both baseline models are presented in the succeeding tables.

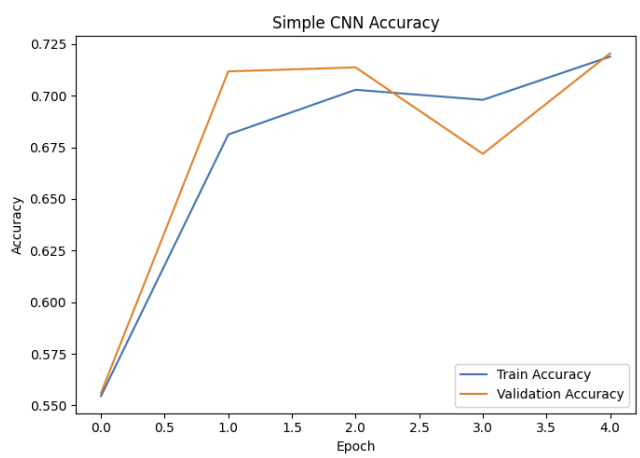

Figure 3. Training and Validation accuracy of the Simple CNN baseline model.

Table 6. Logistic Regression (baseline)

Class	Precision	Recall	F1-Score	Support
Brown Spot	0.71	0.68	0.70	296
Healthy	0.52	0.41	0.46	78
Leaf Blast	0.66	0.64	0.65	361
Rice Hispa	0.79	0.90	0.84	292
Macro avg	0.67	0.66	0.66	1027

Table 7. Simple CNN (baseline)

Class	Precision	Recall	F1-Score	Support
Brown Spot	0.69	0.73	0.71	296
Healthy	0.67	0.03	0.05	78
Leaf Blast	0.64	0.69	0.66	361
Rice Hispa	0.85	0.93	0.89	292
Macro avg	0.71	0.60	0.58	1027

Tables 6 and 7 summarize the per-class precision, recall, and F1-score of the baseline Logistic Regression and Simple CNN models. While both models achieved acceptable performance on some majority classes, their results were less consistent for underrepresented classes, particularly Healthy. The Simple CNN, despite having slightly higher overall accuracy than Logistic Regression, produced a lower macro-F1 score, indicating weaker balanced performance across classes. These findings confirm that the baseline models are useful for comparison, but they are insufficient for reliable multiclass rice leaf disease classification.

Table 8. Accuracy and Macro-F1 of Baseline Models

Model	Accuracy	Macro-F1
Logistic Regression	0.708861	0.662381
Simple CNN	0.720545	0.577858

Table 8 summarizes the performance of the baseline models, including accuracy and macro-F1 scores for Logistic Regression and Simple CNN. These baseline models provide early insights into dataset difficulty. Minority classes, such as Brown Spot and Rice Hispa, tend to have lower macro-F1 scores due to fewer training samples.

7.2 Baseline Deep Learning Model

A baseline deep learning model was implemented to provide early performance benchmarks on the rice leaf disease dataset. Two approaches were considered:

- **EfficientNetB0 (un-tuned)** – a pretrained EfficientNetB0 model with minimal modifications. No hyperparameter tuning was applied, and only minimal augmentation was used.

7.2.1 Training Details:

- Input: 224 x 224 RGB images
- Optimizer: Adam
- Loss: Cross-entropy with class weights
- Epochs: 30
- Augmentation: Flipping, brightness adjustment, contrast adjustment

7.2.2 Initial Results:

The training accuracy and training loss curves of the untuned EfficientNetB0 model during the first phase of training are shown in Figure 4. These curves illustrate the model's early learning behavior, with training accuracy increasing and training loss decreasing across epochs. These trends suggest that the model was able to learn relevant features from the rice leaf images during the initial training stage.

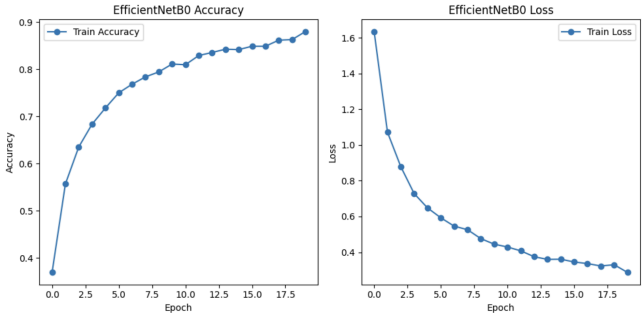

Figure 4. Training accuracy and training loss curves of the untuned EfficientNetB0 model during the first phase of training.

Table 9. Final Model Evaluation Metrics

Metric	Value
Accuracy	0.9883
Macro-F1	0.9827

The final EfficientNetB0 model achieved high performance, obtaining an accuracy of 0.9883 and a macro-F1 score of 0.9827. The close value of these metrics indicates strong and balanced classification across all classes. These results suggest that the model provides reliable and balanced multiclass classification performance for rice leaf disease detection.

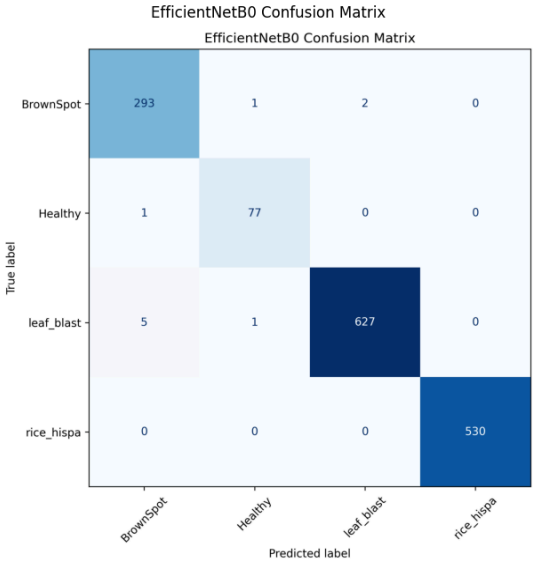

Figure 5. Confusion matrix showing class-wise performance of the EfficientNetB0

Figure 5 presents the confusion matrix of the EfficientNetB0 model on the validation set. This figure visualizes the distribution of correct and incorrect predictions across all rice leaf disease classes, allowing for an assessment of which diseases are most frequently misclassified and identifying classes that may require further attention during subsequent training phases.

7.2.3 Error Analysis (Early Stage):

- Misclassifications occur between visually similar diseases.
- Healthy is the easiest class for the model, while Hispa and Brown Spot are very poorly predicted.
- Healthy class is often misclassified as Leaf Blast.
- Minority classes require careful weighting to improve recall.

7.3 RL Component Stub

A **reinforcement learning (RL)** component was implemented as a stub to demonstrate the framework for future optimization. The environment is defined to interact with the model outputs (predicted class probabilities) and simulate threshold adjustments per class.

The reward function is defined as the change in **macro-F1 score (Δ macro-F1)** after an action, allowing the agent to learn adjustments that improve overall classification performance across all classes. The available actions are to increase or decrease the classification threshold for each disease class.

Initial experiments show stable reward trends across episodes. Figure 6 shows the reward obtained by the Q-learning agent per episode, demonstrating learning progression and validating the interaction between the RL agent and the environment.

Table 10. Reinforcement Learning Threshold Tuning Results

Metric	Value
Base accuracy	0.9935
Base macro-F1	0.9899
Base threshold	0.0
Best threshold	0.6
Best Macro-F1	0.9907
Final policy threshold	0.6
Final policy macro-F1	0.9907

Table 10 summarizes the results of the RL-based threshold tuning. The macro-F1 score improved from 0.9898 at the baseline threshold (0.0) to 0.9907 using the learned threshold of 0.6. The best threshold observed during training was also 0.6, yielding the same macro-F1 score. Accuracy remained high at 0.9934

Although the improvement is modest, the results demonstrate that the RL framework can learn threshold adjustments that slightly enhance balanced classification performance.

7.3.1 Q-Learning Agent

The Q-Learning agent was trained over 50 episodes with 10 steps per episode, exploring 20 discrete threshold values ranging from 0.00 to 0.95. Its goal was to optimize the confidence threshold to improve macro-F1 performance.

As shown in Figure 6, the agent initially received a negative reward but quickly improved and stabilized in subsequent episodes, indicating successful learning and convergence. Minor fluctuations were observed due to exploration, but rewards remained consistently positive in later stages.

Figure 7 shows that macro-F1 scores remained high and stable, ranging from approximately 0.989 to 0.991 throughout training. This indicates that the reinforcement learning process maintained strong classification performance.

Figure 8 illustrates the selected threshold per episode. Early episodes show wide exploration (0.40-0.80), while later episodes converge around 0.55-0.65, indicating that the agent learned a consistent and effective threshold.

Overall, the results demonstrate that the Q-learning agent successfully identified a stable threshold policy while preserving high predictive performance.

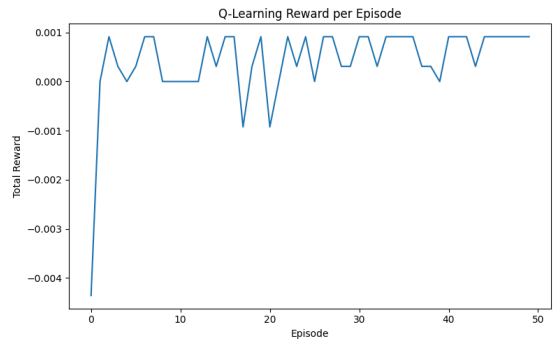

Figure 6. Q-learning reward per episode showing learning progression and convergence behavior.

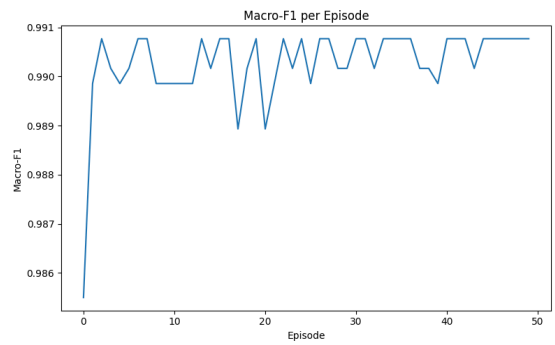

Figure 7. Macro-F1 score per episode demonstrating stable classification performance during reinforcement learning.

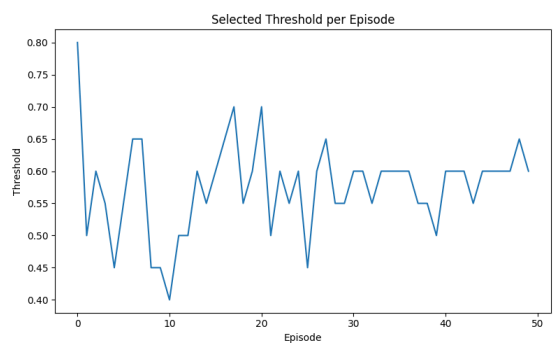

Figure 8. Selected threshold per episode illustrating exploration and convergence toward an optimal decision threshold.

8. Grad-CAM Visualization Analysis

A Grad-CAM (Gradient-weighted Class Activation Mapping) component was implemented to provide visual explanations of the model's predictions. This component generated heatmaps that highlight the regions of the input image that most influenced the classification decision. By overlaying these heatmaps on the original rice leaf images, the model's focus areas can be inspected and validated.

Figure 7 shows a sample Grad-CAM visualization for a Brown Spot leaf image. The heatmap highlights the areas of the leaf where brown lesions are present, which are the key characteristics of the disease. The model correctly focuses on these affected regions rather than the background, indicating that it is learning relevant visual features. The high confidence score of 99.99% also supports that the prediction is accurate.

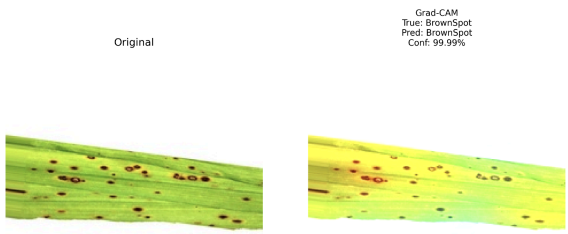

Figure 7. Grad-CAM visualization highlighting model attention on the Brown Spot leaf image.

This result shows that the model is able to identify important disease-related features such as lesion patterns and discoloration. The Grad-CAM component helps verify that the model's predictions are based on meaningful parts of the image, improving interpretability and user trust in the system.

9. Ablation Experiments

Ablation experiments were conducted to evaluate the contribution of key components in the model pipeline. Specifically, two experiments were performed: (1) removal of data augmentation and (2) removal of class weighting. These modifications were applied independently to assess their impact on model performance using accuracy and macro-F1 score. The results provide insight into how each component contributes to generalization and balanced classification performance.

Table 11. Ablation Study Results

Experiment	Accuracy	Macro-F1
EfficientNetB0 model	0.9935	0.9899
No Augmentation	0.9928	0.9880
No Class Weights	0.9915	0.9792

The ablation study evaluates the contribution of key components in the model by systematically removing them and observing the resulting performance changes. The EfficientNetB0 model achieved the highest performance in terms of accuracy and macro-F1 score. Removing data augmentation resulted in a slight decrease in accuracy (-0.0007) and macro-F1 (-0.0019), indicating that augmentation contributes to improved generalization by enabling the model to handle variations such as lighting, orientation, and noise. However, the relatively small performance drop suggests that the dataset already contains consistent visual patterns. In contrast, removing class weights led to a more significant decline in macro-F1 score (-0.0107) compared to accuracy (-0.002), showing that the model became biased toward majority classes. This imbalance particularly affected the minority class (Healthy), where recall dropped to 0.88, demonstrating that accuracy alone does not fully capture performance degradation and that macro-F1 is a more reliable metric for evaluating balanced classification.

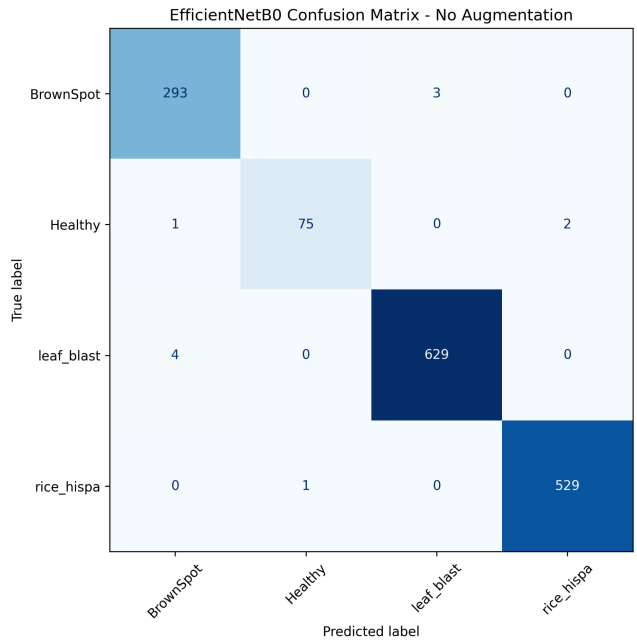

Figure 8. Confusion matrix of the EfficientNetB0 model without data augmentation.

Figure 8 shows the confusion matrix of the model trained without data augmentation, where overall classification performance remains high but with a slight increase in misclassification compared to the baseline model. The errors are primarily observed between visually similar classes, such as Brown Spot and Leaf Blast, which share similar lesion characteristics. Without augmentation, the model is exposed to fewer variations during training, making it less robust to changes in brightness, orientation, and background conditions. This confirms that augmentation improves generalization by helping the model better handle variations in real-world scenarios, even if its impact on overall performance metrics is relatively small.

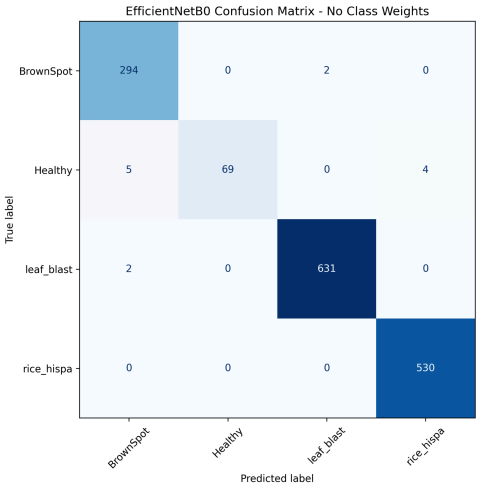

Figure 9. Confusion matrix of the EfficientNetB0 model without class weighting.

Figure 9 presents the confusion matrix of the model trained without class weighting, where more pronounced misclassifications occur, particularly affecting the minority class (Healthy). The model tends to favor majority classes such as Leaf Blast and Brown Spot, leading to reduced recall and misclassification of Healthy samples, especially when leaf textures resemble early disease symptoms. This imbalance results in a lower macro-F1 score despite relatively high accuracy, demonstrating that the model prioritized frequently occurring classes when class weights are not applied. These findings confirm that class weighting is essential for mitigating bias and ensuring fair performance across all disease categories.

The ablation experiments demonstrate that both data augmentation and class weighting contribute to model performance, but their impact differs in magnitude. Data augmentation provides incremental improvements by enhancing generalization and robustness to variations in input images, while class weighting plays a more critical role in addressing class imbalance and ensuring fair performance across all classes. The significant drop in macro-F1 score when class weights are removed highlights its importance in maintaining balanced classification, particularly for minority classes such as Healthy. These findings further validate the importance of evaluating models using macro-F1 score in imbalanced classification tasks, as it provides a more reliable measure of performance across all classes compared to accuracy alone.

9.1 Error and Slice Analysis

Error and slice analysis were conducted to further interpret the results of the ablation experiments and understand model behavior at a class level. The analysis shows that most misclassifications occur between visually similar disease classes, particularly Brown Spot and Leaf Blast, due to overlapping lesion patterns. In addition, Healthy samples are occasionally misclassified when leaf textures resemble early disease symptoms. Slice analysis reveals that minority classes, especially Healthy, experience reduced recall when class weights are removed, confirming the model's bias toward majority classes under imbalance conditions. These findings indicate that the observed performance drops in the ablation study are not random but are driven by class imbalance and visual similarity between disease patterns. Furthermore, Grad-CAM visualizations support this observation by showing that the model still focuses on meaningful leaf regions even in incorrect predictions, suggesting that errors arise from feature similarity rather than irrelevant attention.

10. Limitations

Despite the strong validation performance, several limitations must be acknowledged. First, the dataset consists of images captured under controlled conditions with clean, uniform backgrounds, which may not fully represent the visual diversity of real paddy field environments. Accordingly, performance may decrease when the model encounters images with soil clutter, overlapping leaves, moisture variation, or mixed lighting conditions. Second, the evaluation was conducted entirely on a held-out validation split; a dedicated test set was not maintained separately throughout the experiment, which limits the ability to make fully independent performance claims. Third, the RL threshold tuning was performed on the same validation set used for model selection, which introduces a degree of optimistic bias in the reported threshold improvement. Fourth, the knowledge base used by the NLP component is small (four entries) and manually authored, meaning the quality of retrieved explanations is limited by the breadth of the original descriptions. Fifth, all experiments were conducted without GPU acceleration, which constrained the number of architectures and hyperparameter configurations that could be explored.

11. Conclusion

This study presented a rice leaf disease detection system using a fine-tuned EfficientNetB0 convolutional neural network for the classification of four rice leaf conditions prevalent in Philippine agriculture: Brown Spot, Leaf Blast, Rice Hispa, and Healthy. The proposed system achieved an accuracy of 0.9935 and a macro-F1 score of 0.9899 on the validation set, substantially outperforming the logistic regression baseline (accuracy: 0.7089, macro-F1: 0.6624) and the simple CNN baseline (accuracy: 0.7205, macro-F1: 0.5779).

Ablation experiments demonstrated that both data augmentation and class weighting contribute to model performance, with class weighting having a more pronounced effect on balanced classification. Removing class weight resulted in a macro-F1 drop of 0.0107, disproportionately affecting the minority Healthy class, while removing augmentation caused a smaller decline of 0.0019. These findings reinforce the importance of using macro-F1 as the primary evaluation metric in imbalance classification tasks. Grad-CAM visualizations confirmed that the model attends to disease-relevant leaf regions rather than background noise, supporting the interpretability of predictions for end users. The auxiliary NLP component successfully retrieved contextually appropriate disease descriptions using TF-IDF similarity matching, providing users with actionable explanations alongside each prediction. The Q-learning reinforcement learning agent further improved the macro-F1 score from 0.9899 to 0.9907 by learning an optimal confidence threshold of 0.60 over 50 training episodes.

The system is designed as a decision support tool for Filipino farmers and agricultural technicians, supplementing field expertise rather than replacing it. Future work should focus on expanding the dataset to include field-collected images from Pampanga and other Philippine rice-growing regions, integrating treatment recommendation modules, and evaluating model performance on mobile or edge deployment platforms, and

extending classification to additional rice diseases common in Southeast Asian paddy environments.

12. ETHICS RISK REGISTER

12.1 Misdiagnosis Leading to Crop Loss

CNN classification errors may lead to misdiagnoses, which could contribute to inappropriate disease management decisions and potentially affect crop yield [3]. To mitigate this, the system displays confidence scores and Grad-CAM heatmaps for every prediction. All system outputs carry a clear disclaimer encouraging users to confirm any treatment decision with a licensed agricultural technician or extension officer. The system is intended as a decision support tool and is not designed for fully automated agricultural or clinical decision-making. All predictions should be validated by agricultural experts before applying any treatment.

12.2 Dataset Bias from Class Imbalance

Some disease classes may be underrepresented in publicly available datasets, which can reduce performance on minority classes [8]. To mitigate this risk, class weights are applied during model training so that underrepresented classes contribute proportionally more to the loss function. Model evaluation uses macro-F1 score and per-class confusion matrices to evaluate performance and ensure fairness across all classes.

12.3 Overreliance on Automated Recommendations

Users may depend too heavily on the system's output without considering specific field conditions or unique farm situations. The system is meant to support decision-making, not replace expert judgment.

12.4 Dataset Representativeness Limitations

The RiceLeafs dataset consists of images captured in controlled environments, with clean backgrounds, consistent lighting, and minimal field noise. It is also not region-specific, meaning it may not fully represent visual conditions in Pampanga or other Philippine rice fields. As a result, real-world images containing soil, clutter, moisture, overlapping leaves, or mixed lighting may produce reduced accuracy. Hence, augmentation strategies simulate field-like variation (brightness shifts, contrast changes, random flips).

12.5 Transparency and Responsible Use

Because machine learning models operate as complex black boxes, users may misinterpret or overtrust system outputs. Transparent explanations are essential to avoid misuse and ensure informed decision-making. Users are provided with responsible-use guidelines emphasizing that all predictions are preliminary and must be corroborated with field expertise.

13. TEAM ROLES AND MILESTONES

13.1 Team Roles

- Francine A. G. Alejandro - Project & Integration Lead: Defines scope, manages timeline, oversees component integration, handles release tagging, coordinates defense.
- Yohanna A. Decilio - Data & Ethics Lead: Sources datasets, manages licensing and consent, conducts bias checks, writes ethics statements.
- Shane S. Mendoza - Modeling Lead (CNN/NLP): Designs models, develops training pipelines, conducts ablation studies, performs error analysis, prepares evaluation metrics, and reports.
- Maria N. G. L. Pascua - Documentation & MLOps Lead: Sets up repository, tracks issues, maintains project board, ensures reproducibility, prepares environment files, and automates scripts/CI.

13.2 Milestones

Week 1

- Project kickoff; team roles assigned
- Dataset exploration; initial data acquisition
- Repository setup; GitHub board created
- Initial CNN and NLP research; literature review

Week 2

- Data acquired and cleaned; train/validation/test splits finalized
- Baseline models trained (simple ML and DL); initial metrics logged
- CNN experiment running with first results; NLP component scaffolded/prototyped
- RL agent stubbed with reward design; early learning curves prepared
- Draft model card sections started; Ethics statement updated
- Deliverable: 1-2 page Checkpoint Report and updated repository

Week 3

- Code frozen and reproducible
- Results tables/plots generated; ablations (≥2); error/slice analysis completed
- Final Report (4-6 pages) in conference-style PDF format
- Video Presentation with recorded demonstration

14. REFERENCES

[1] Philippine Statistics Authority. 2025. Agriculture and Fisheries Indicators System: Tracking Economic Growth, 2020–2024. Agriculture and Fisheries Indicators System. *Psa.gov.ph*. Retrieved from https://psa.gov.ph/system/files/main-publication/AFIS%20Economic%20Growth%20Agriculture%202020-2024_signed.pdf

[2] Philippine Statistics Authority. 2023. Crops Production Survey 2020. *psada.psa.gov.ph*. Retrieved from https://psada.psa.gov.ph/catalog/234

[3] Chinna Gopi Simhadri, Hari Kishan Kondaveeti, Valli Kumari Vatsavayi, Alakananda Mitra, and Preethi Ananthachari. 2024. Deep learning for rice leaf disease detection: A systematic literature review on emerging trends, methodologies, and techniques. Information processing in agriculture (May 2024). DOI:https://doi.org/10.1016/j.inpa.2024.04.006

[4] Sharada P. Mohanty, David P. Hughes, and Marcel Salathé. 2016. Using Deep Learning for Image-Based Plant Disease Detection. *Frontiers in Plant Science* 7, (September 2016). DOI:https://doi.org/10.3389/fpls.2016.01419

[5] Wallace F. L. Chin and Wida S Suhaili. 2025. Class-Specific Cross Attention for Fine-Grained Paddy Disease Classification. *2025 7th International Conference on Applied Computational Intelligence in Information Systems (ACIIS)* (October 2025), 1–6. DOI:https://doi.org/10.1109/aciis66255.2025.11402958

[6] Mingxing Tan and Quoc V Le. 2019. EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks. *arXiv.org*. Retrieved from https://arxiv.org/abs/1905.11946

[7] Alhassan Mumuni and Fuseini Mumuni. 2022. Data augmentation: A comprehensive survey of modern approaches. *Array* 16, (November 2022), 100258. DOI:https://doi.org/10.1016/j.array.2022.100258

[8] Mateusz Buda, Atsuto Maki, and Maciej A. Mazurowski. 2018. A systematic study of the class imbalance problem in convolutional neural networks. *Neural Networks* 106, (October 2018), 249–259. DOI:https://doi.org/10.1016/j.neunet.2018.07.011

[9] Shahariar Sarkar. 2025. From Black-Box to Transparent AI in Agriculture: Integrating Grad-CAM and Counterfactual Explanations for Rice Leaf Disease Detection. *2025 7th International Conference on Electrical Information and Communication Technology (EICT)* (December 2025), 1–6. DOI:https://doi.org/10.1109/eict68394.2025.11355586

[10] R Lokesh. "Rice Leaf Diseases Detection." *www.kaggle.com*, 2023, www.kaggle.com/datasets/loki4514/rice-leaf-diseases-detection.

[11] Ramprasaath R. Selvaraju, Michael Cogswell, Abhishek Das, Ramakrishna Vedantam, Devi Parikh, and Dhruv Batra. 2017. Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization. *2017 IEEE International Conference on Computer Vision (ICCV)* (October 2017), 618–626. DOI:https://doi.org/10.1109/iccv.2017.74

[12] Richard S Sutton and Andrew Barto. 2018. *Reinforcement learning: An introduction* (2nd ed.). The Mit Press, Cambridge, Ma ; London.